

Effects of backrest design on biomechanics and comfort during seated work

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Abstract

The purpose of this study was to examine the effects of backrest configuration on seatpan and backrest pressure, spinal posture, and comfort. Thirty volunteers (15 male, 15 female) typed a standardized text passage while seated at a computer workstation in five backrest configurations: chair only, chair with a supplementary backrest, and with each of three lumbar pad thicknesses. Pressure, lumbar and cervical angles were collected during 15-min trials. Subjective data were collected during each trial and at the end of the entire protocol. The addition of a supplementary backrest to a standard chair reduced peak and average pressure on the back by 35% and 20%, respectively ($P < 0.02$). Lumbar lordosis was observed only when lumbar pads were used, being greatest with the large pad. Participants preferred backrest configurations that had lower pressure on the back and less lordotic lumbar posture (backrest only or 3 cm lumbar pad), regardless of anthropometrics. Comfort was rated highest in conditions that would not necessarily be considered biomechanically ideal. Further delineation between specific comfort and objective seating variables is required to effectively reduce and prevent low back pain.

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1. Introduction

Over the past century, there has been an increase in the proportion of seated occupations (van Dieen et al., 2001) and although these jobs are generally less physically demanding, seated work is a risk factor in the development of low back pain (Cartas et al., 1993; Lingsfeld et al., 2000; Videman et al., 1990). Substantial health costs and the increase in sedentary occupations have led to a considerable amount of attention directed toward office seating, as the chair industry represents 25% of a \$10 billion a year office furniture industry in the United States (Freeland, 1998). Amongst other aspects in seating and chair design, the backrest has become a focal point.

The role of a chair backrest is to attenuate the stresses exerted on the vertebral column by relaxing the erector spinae musculature, while maintaining lumbar lordosis and increasing comfort (Corlett and Eklund, 1984). Backrests have been evaluated using spine posture (Andersson et al., 1979; Bendix et al., 1996; Bishu et al., 1991; Dolan et al., 1988; Lord et al., 1997; Makhosous et al., 2003; Vergara and Page, 2000b), pressure measures (Dunk and Callaghan, 2005), electromyography (EMG) (Andersson et al., 1974; Andersson and Ortengren, 1974; Dolan et al., 1988; Makhosous et al., 2003; van Dieen et al., 2001), and subjective measures, such as discomfort scales and questionnaires (Bishu et al., 1991; Coleman et al., 1998; Corlett and Bishop, 1976; Shackel et al., 1969; van Dieen et al., 2001; Vergara and Page, 2000a, 2002).

While comfort is an important criterion to the user, there has been little research to relate comfort to biomechanical variables, such as pressure or low back posture in office seating. Seatpan and backrest pressures have been used in

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the evaluation of car seats (Andreoni et al., 2002; Gyi and Porter, 1999; Porter et al., 2003) and wheelchairs (Houle, 1969; Parent et al., 2000; Sprigle et al., 1990; Takechi and Tokuhito, 1998), for which comfort and function are very important. Lumbar spine posture has been linked to low back pain in seating (Dolan et al., 1988; Wilder and Pope, 1996). However, posture measurements when seated typically require alterations of the backrest and used cumbersome equipment, both of which likely alter natural behaviour (Bendix et al., 1996; Bishu et al., 1991; Parent et al., 2000). Vergara and Page (2000a,b) developed two simple devices which did not require alteration of the backrest or interfere with the user. Most interestingly, they proposed a “fidgeting” index, which assessed small movements using lumbar and pelvic angle amplitudes and frequencies (Vergara and Page, 2002). Overall, the relationships between comfort and objective measures are not clearly defined. Pressure distribution appears to be related to comfort in car seats, but has not been shown in office applications, while posture and muscle activity are even less clear (de Looze et al., 2003).

With the large selection of consumer products available, there is a need to establish backrest design and selection criteria based on relationships between comfort, pressure, posture, and anthropometrics. The purpose of this paper was to examine the effects of backrest configuration on backrest and seatpan interface pressures, lumbar and cervical spine posture, and comfort.

2. Methods

2.1. Participants

Thirty adult volunteers (15 male, 15 female) with no history of low back pain were recruited from the University community. Participation required the ability to type continuously at a minimum rate of 20 words/min. Anthropometric data including, height, weight, hip girth, waist girth, leg length (greater trochanter to lateral malleolus), torso length (greater trochanter to C7), and shoulder (biacromial) breadth measurements were collected. These data were used to calculate body mass index (BMI), waist–hip ratio (WHR), and leg–torso ratio (LTR).

2.2. Protocol

Participants typed a standardized text passage at a computer workstation using a non-adjustable chair without armrests or wheels, in five randomized backrest conditions: (i) chair only (CO); (ii) chair and supplementary backrest only (BO); (iii) chair and supplementary backrest with small lumbar pad (SPAD); (iv) chair and supplementary backrest with medium lumbar pad (MPAD); (v) chair and supplementary backrest with large lumbar pad (LPAD). The supplementary backrest used in this study has a polyurethane frame with a foam outer layer (ObusForme Lowback Backrest Support, ObusForme Ltd., Toronto, ON). It is

supplied with a removable lumbar pad (6 cm thick) that is secured by Velcro. In addition to the standard lumbar pad, small (3 cm thick) and large (9 cm thick) pads were created as 50% and 150% of the standard (medium) pad thickness, respectively (Fig. 1). The location of the lumbar pad was standardized by aligning the most prominent part of the lumbar pad at the mid-lumbar level (L3). The seatpan was moved forward 5 cm to allow for the thickness of the supplementary backrest.

The workstation was adjusted for each participant according to the Canadian Standards Association (CSA) guidelines on office ergonomics (CSA-Z412-00, 2000). One individual required the use of a footrest to meet guidelines. Armrests and wheels were not used as both had the potential to be a confounding variable and complicated the assessment of pressure. Each of the five backrest conditions was tested for 15 min, separated by 5-min breaks, during which participants were allowed to move about as desired. To allow collection of both subjective and objective measures at set intervals, as well as to ensure that the measures were not interrupted by the protocol, specific intervals were used for collection and analysis of data. Seatpan and backrest pressure, as well as lumbar and cervical angles were measured over 5-min periods, beginning at the onset and 10th minute of each trial. Within each 5-min collection period, three 1-min samples were collected beginning at 0, 2, and 4 min (Fig. 2). Participants were informed that data would be collected intermittently throughout the trial, but were not told when the collection occurred. Three subjective measures were administered. First, a computer-based Body-Part Discomfort Questionnaire (BPDQ, adapted from Corlett and Bishop, 1976) was completed at the 5th, 9th and 15th minute of each condition so as to not interfere with pressure and angle data collection. The BPDQ consisted of a 13-body-part diagram of an individual on which participants indicated their level of discomfort based on a 5-point scale ranging from 0 (no discomfort) to 5 (very uncomfortable). Second, a Backrest Condition Rating (BCR) scale in which participants rated the backrest condition by placing a mark along a 5-point Likert Scale, ranging from ‘Dislike Immensely’ to ‘Like Immensely’, was completed at the end of each backrest condition (15th minute). Finally, at the end of the entire protocol, each backrest condition was ranked from “Best” (1) to “Worst” (5) and participants expressed any comments regarding each condition. Each trial was also photographed in the sagittal plane.

2.3. Backrest and seatpan pressure

A capacitive pressure-sensing system consisting of two 36×36 half inch element arrays ($46 \text{ cm} \times 46 \text{ cm}$) was used to measure the pressure on the backrest and the seatpan (X236, Xsensor Technology Corporation, Calgary, AB). The pressure mats were leased from the manufacturer and calibrated prior to arrival with application of the calibration data prior to each collection. Pressure data were sampled at 3 Hz

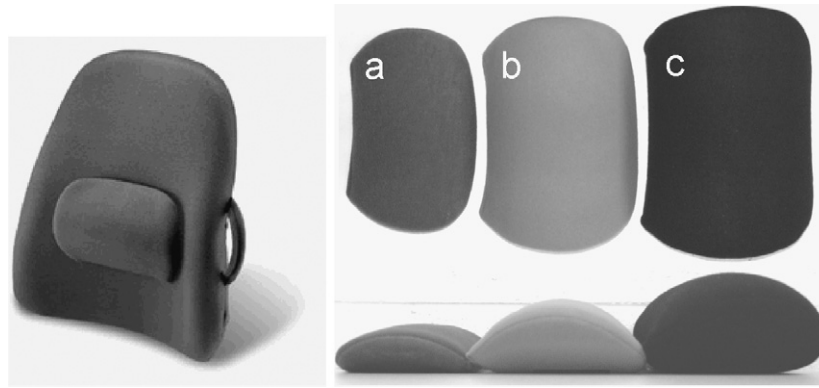


Fig. 1. Supplementary backrest and lumbar pads. (a) Small pad (–50% of medium pad thickness), (b) medium pad, and (c) large pad (+50% of medium pad thickness).

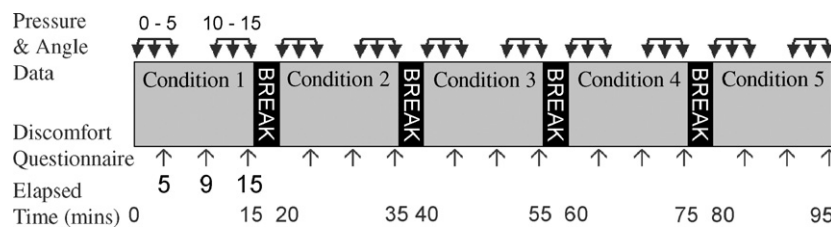


Fig. 2. Collection protocol. The 5 randomized backrest conditions were each tested for 15 min in duration with 5-min break periods in between. For each condition, 3 1-min samples of pressure and angle data were collected over 5-min periods at 0 and 10 min. Body-part discomfort questionnaires were completed at 5, 9 and 15 min, with a backrest condition rating completed at the end of each backrest condition (15th minute).

using a pressure range of 5–200 mmHg. System reported values included the mean of the average pressure (AP), mean peak pressure (MPP), and mean contact area (MCA) over the duration sampled. In addition, raw pressure data were used to calculate the centre of pressure (CP) for the seatpan and the backrest for each sample. CP was calculated as the centroid, or the weighted average, of all pressure intensities and locations. For the seatpan, the anterior–posterior coordinates for the CP (CP_{S-AP}) increased from the front edge (0 cm) to the rear (46 cm) of the seatpan and the medial–lateral CP (CP_{S-ML}) increased from the right (0 cm) to the left (46 cm) side of the seatpan. For the backrest, the height of the CP (CP_{B-H}) increased from zero at the bottom of the backrest to a maximum of 46 cm at the top and the medial–lateral CP (CP_{B-ML}) was defined the same as the seatpan. Due to the changing size of the lumbar pad, the top of the mat was not always at the same height (i.e. it was lower as the pad thickness increased). To assess the frame to frame variability of the CP over each trial, two additional measures were used: (i) the standard deviation of the CP over the trial (CP variability) and (ii) CP excursion in absolute terms.

2.4. Lumbar and cervical angles

Lumbar and cervical angles were assessed in the sagittal plane using a custom-made system, each consisting of a flexible, unidirectional sensor for the lumbar and cervical areas of the spine (11.4 cm × 0.64 cm × 0.05 cm, Flex Sensor, Images SI Inc., Staten Island, NY). Each sensor was placed in a fabric sleeve which was affixed to the lower

back and neck using two elastic straps at either end of the sleeve. The sensors were centred over the spinous processes in the lumbar (L3) and cervical (C4) regions. For both the lumbar and cervical sensor, the upper strap held the sensor in place (the upper strap for the cervical sensor was positioned above the ears and around the forehead), while the lower strap served as a slip channel to allow the sensor to glide up and down over the spinous processes with flexion and extension (Fig. 3).

The sensors were calibrated prior to being mounted on the participant using fixed angles of 0° and 30°, (linearity confirmed with 60°). Reference cervical and lumbar angles were measured in both natural standing and erect sitting postures. Lumbar angles were also collected while the participant moved from maximal lordosis to a straight lumbar spine in both standing and sitting postures. Similarly, cervical angles were collected by instructing the participant to begin with the neck fully flexed and to slowly extend until they were at a comfortable end range in both standing and sitting postures. Angle data were collected at 50 Hz using custom software (LabVIEW, National Instruments Corp., Austin, TX). Spine angles were reported in degrees and as a difference from natural lordosis when standing (lumbar region) and as a difference of the natural erect sitting posture (cervical).

2.5. Statistical analyses

Data were analysed using a 2 (gender) × 5 (backrest condition) mixed analysis of variance (ANOVA) for each

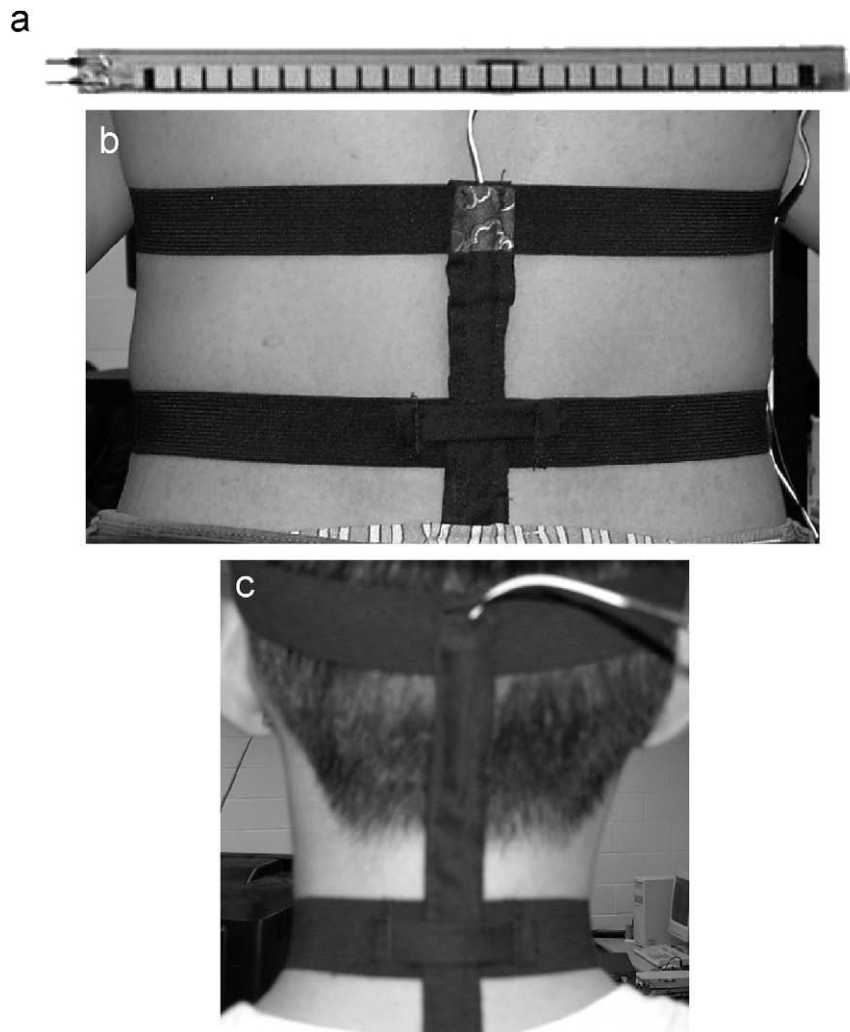


Fig. 3. Sensor set-up. Custom-made system consisting of two flexible sensors (a–1 shown), each placed in a fabric sleeve and affixed to the lower back (b) and neck (c) via two elastic straps. The sensors were centred over the spinous processes in the L3 region (b) and C4 region (c) of the spine.

variable, where gender was the between-groups factor and the backrest condition was the repeated measure, with an alpha level of 0.01. Due to the potential number of follow-up contrasts, the desired p -value of 0.05 was adjusted using a modified Bonferroni correction to reduce the chance of false positives resulting in an effective p -value of 0.02 for all follow-up contrasts. Multiple regression analyses were also performed to determine relationships between selected variables, as well as to find predictors (biomechanical or anthropometric) of comfort and backrest configuration preference. For multiple regression models, an alpha level of 0.05 was used. All analyses were performed using Statistica 6.0 (StatSoft Inc., Tulsa, OK).

3. Results

3.1. General

Participant anthropometric and age data are summarized in Table 1. There were no changes in pressure or angle

data with respect to time within any backrest condition, thus objective data were averaged over each trial.

3.2. Backrest pressure variables

3.2.1. Average pressure (AP)

Average backrest pressure (AP) was significantly affected by the backrest condition [$F_{4,112} = 158.19$, $p = 0.00001$]. The BO condition yielded the lowest pressure, followed by CO, SPAD, MPAD and LPAD (Table 2 and Fig. 4(a)). Contrast analyses revealed that all conditions differed significantly ($p < 0.02$). The addition of a supplementary backrest (BO) significantly reduced AP on the backrest by 20% compared to the CO (16 ± 0.3 versus 20 ± 0.5 mmHg, respectively).

3.2.2. Mean peak pressure (MPP)

Analysis of the MPP revealed a significant effect of backrest condition [$F_{4,112} = 7.63$, $p = 0.00002$] (Table 2). The BO condition had significantly lower backrest MPP

Table 1
Participant data and mean anthropometric measures for females and males

Gender	Age (year)	H (m)	W (kg)	SB (m)	WG (m)	HG (m)	LL (m)	TL (m)	BMI (kg/m ²)	LTR	WHR
F	21.1 (2.2)	1.65 (0.05)	60.5 (9.0)	0.36 (0.02)	0.74 (0.07)	0.97 (0.06)	0.78 (0.03)	0.57 (0.05)	22.2 (3.3)	1.38 (0.14)	0.76 (0.04)
M	24.7 (2.9)	1.77 (0.08)	79.0 (14.8)	0.41 (0.03)	0.86 (0.11)	1.00 (0.08)	0.83 (0.05)	0.61 (0.04)	25.3 (4.2)	1.37 (0.07)	0.86 (0.06)
MEAN	23.0 (3.1)	1.71 (0.09)	69.7 (15.3)	0.39 (0.03)	0.80 (0.11)	0.98 (0.07)	0.81 (0.05)	0.59 (0.05)	23.70 (4.0)	1.38 (0.11)	0.81 (0.07)

Values in parentheses represent standard deviation. H = height; W = weight; SB = shoulder breadth; WG = waist girth; HG = hip girth; LL = leg length; TL = torso length; BMI = body mass index; LTR = leg length–torso length ratio; WHR = waist girth–hip girth ratio.

Table 2
Pressure variables (values in parentheses indicate standard error)

Variable	Backrest condition				
	CO Mean	BO Mean	SPAD Mean	MPAD Mean	LPAD Mean
Seatpan					
Average pressure (mmHg)	43.0 ^a (.8)	43.4 ^a (.7)	43.0 ^a (.7)	43.3 ^a (.8)	43.5 ^a (.8)
Mean contact area (cm ²)	1130.8 ^a (23.2)	1151.8 ^{ab} (24.1)	1166.7 ^b (22.2)	1145.2 ^a (22.9)	1130.7 ^a (21.5)
Centre of pressure _{S-AP} (cm)	30.7 ^a (0.3)	30.7 ^a (0.3)	30.3 ^a (0.3)	29.2 ^b (0.3)	28.3 ^c (0.3)
Centre of pressure _{S-ML} (cm)	25.0 ^b (0.1)	24.7 ^a (0.1)	24.6 ^a (0.1)	24.6 ^a (0.2)	24.8 ^a (0.2)
Backrest					
Average pressure (mmHg)	20.4 ^a (0.5)	16.3 ^b (0.3)	25.5 ^c (0.8)	27.2 ^d (0.6)	29.0 ^c (0.6)
Mean peak pressure (mmHg)	53.6 ^a (5.5)	34.7 ^b (0.9)	43.9 ^c (1.6)	47 ^c (1.4)	51.6 ^{ad} (2.0)
Mean contact area (cm ²)	413.8 ^a (21.1)	329.6 ^{bc} (19.8)	327.2 ^b (13.2)	354.7 ^c (11.3)	373.0 ^a (10.1)
Centre of pressure _{B-H} (cm)	27.5 ^a (0.7)	29.1 ^a (0.9)	28.3 ^a (0.5)	28.5 ^a (0.4)	28.7 ^a (0.4)
Centre of pressure _{B-ML} (cm)	26.3 ^a (0.3)	25.7 ^a (0.5)	24.6 ^b (0.3)	24.8 ^b (0.2)	24.5 ^b (0.2)

Means with different letters represent significant difference at $p < 0.02$.

compared to all other conditions ($p < 0.02$). The addition of a supplementary backrest significantly lowered the backrest MPP by 35% compared to the chair alone ($p < 0.02$) (Fig. 4(b)). SPAD was not significantly lower than MPAD, but both were significantly lower than CO and LPAD, while LPAD produced significantly higher backrest MPP than all other conditions ($p < 0.02$).

3.2.3. Mean contact area (MCA)

The mixed ANOVA for the MCA on the backrest revealed a significant interaction between gender and backrest condition [$F_{4,112} = 4.89$, $p = 0.001$], which was due to a differential influence of the addition of the small pad on males and females (Table 2). Specifically, the MCA decreased for males when moving from BO to SPAD, while it increased for females. Males had significantly greater backrest contact area ($p < 0.02$) than females in all conditions. In addition, there was a main effect of backrest condition on MCA [$F_{4,112} = 13.81$, $p < 0.000001$]. CO had the largest contact area, followed by LPAD, MPAD, BO, with SPAD having the smallest MCA (Fig. 4(c)). CO had a significantly larger contact area than all conditions except LPAD ($p < 0.02$), while BO and SPAD MCAs were significantly smaller than CO by approximately 20%. All lumbar pad conditions (SPAD, MPAD and LPAD) differed significantly ($p < 0.02$). Qualitatively, the lumbar pads isolated pressure to the lumbar region while pressure

was predominantly in the thoracic region without a lumbar pad (CO and BO).

3.2.4. Backrest CP

Backrest condition approached significance with respect to the CP height (CP_{B-H}) [$F_{4,108} = 3.2$, $p = 0.02$]. CP_{B-H} was lowest for CO followed by SPAD, MPAD, LPAD and BO, with the vertical CP ranging 27.5–29.1 cm above the seatpan (Table 2). The medial–lateral CP on the backrest (CP_{B-ML}) was affected by backrest condition [$F_{4,108} = 16.6$, $p = 0.000001$] and ranged from 24.5–26.3 cm.

All measures of backrest CP variability had similar relationships across conditions and each attained statistical significance (height, medial–lateral, and excursion, all $F_{4,108} > 12.3$, all $p < 0.00001$). BO consistently produced the highest variability for most variables, while LPAD had the lowest variability across almost all conditions for each variable except for CPV_{B-ML} (Fig. 4(f)).

3.3. Seatpan pressure variables

3.3.1. Average pressure (AP)

The mixed ANOVA conducted on the average seatpan pressure data revealed a significant gender effect [$F_{1,28} = 28.0$, $p = 0.00001$]. Males had significantly higher average seatpan pressure (47.0 ± 1.0 mmHg) compared to

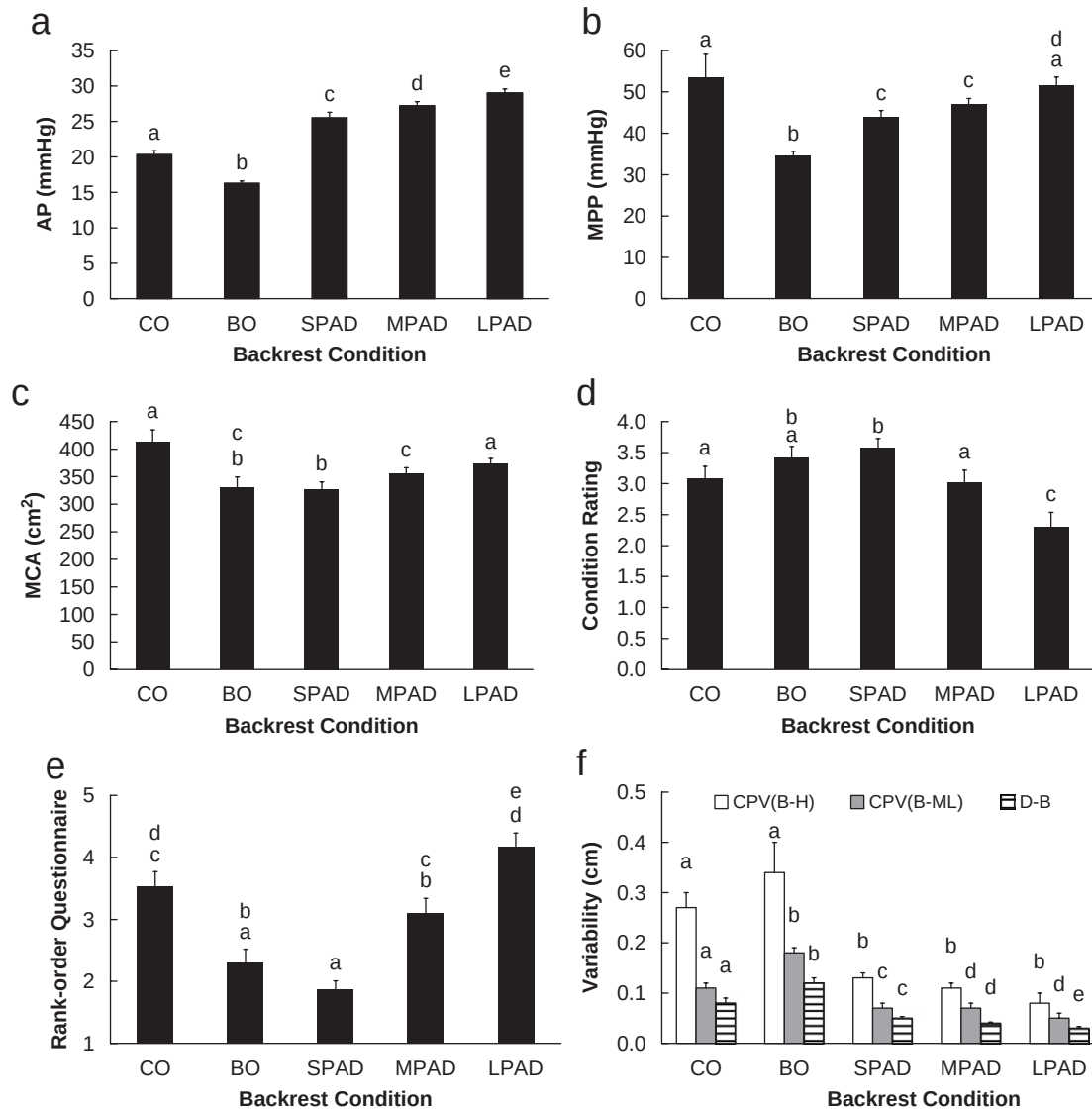


Fig. 4. Backrest pressure and subjective data as a function of backrest condition (with standard error, gender pooled). (a) average pressure, (b) mean peak pressure, (c) mean contact area, (d) backrest condition rating, (e) rank-order questionnaire and (f) backrest CP variability (CPV(B-H) = backrest height, CPV(B-ML) = medial lateral, D-B = mean excursion). Bars with different letters represent significant difference at $p < 0.02$.

females (39.5 ± 1.0 mmHg). No effect of backrest condition was found (Table 2).

3.3.2. Mean peak pressure (MPP)

The MPP and variability for the seatpan was similar for each backrest condition, as they all reached the maximum threshold of the pressure mat (200 mmHg) in at least one array element of the mat.

3.3.3. Mean contact area (MCA)

The mixed ANOVA for mean seatpan contact area revealed a significant main effect of backrest condition [$F_{4,112} = 5.3$, $p = 0.0006$] (Table 2). The largest contact area on the seatpan was found with SPAD, which was significantly greater than all other conditions ($p < 0.02$) except for the backrest alone (Table 2). However, when

controlling for other variables (LTR, BMI, backrest contact area, and lumbar angle), multiple regression analysis indicated no relationship between backrest condition and seatpan contact area.

3.3.4. Seatpan CP

Backrest condition significantly altered the anterior-posterior seatpan CP (CP_{S-AP}) [$F_{4,108} = 64.8$, $p = 0.00001$] with CO resulting in a significantly more posterior CP_{S-AP} location (30.7 ± 0.3 cm) than all others except BO (30.7 ± 0.3 cm) and SPAD (30.3 ± 0.3 cm) ($p < 0.02$) (Table 2). The LPAD condition (28.3 ± 0.3 cm) was significantly closer to the front of the seatpan than all other conditions ($p < 0.02$). Medial-lateral CP on the seatpan (CP_{S-ML}) was also significantly affected by backrest condition [$F_{4,108} = 3.8$, $p = 0.006$]. The CP_{S-ML} for the CO condition was shifted

more to the left than the SPAD, MPAD and LPAD conditions ($p < 0.02$) (Table 2). Across all conditions, there was a mean shift of 1.7 ± 0.2 cm to the left of the seatpan centre (23 cm).

3.4. Spine posture

3.4.1. Lumbar angle

Lumbar angles and the normalization process were highly variable. The mean natural standing lumbar angle was measured to be $12.8 \pm 1.9^\circ$. A significant main effect of backrest condition was found for lumbar angle [$F_{4,104} = 46.8$, $p = 0.00001$] ($N = 28$ due to two missing data points). The difference in the lumbar angle between each condition was significant ($p < 0.02$) with a successive increase for each lumbar pad condition, from small to large (Fig. 5). Both the supplementary backrest alone (BO) and the chair alone resulted in straight or slightly flexed lumbar spine postures ($16.3 \pm 1.9^\circ$ and $14.2 \pm 1.8^\circ$, respectively) (Fig. 5). The addition of the small lumbar pad decreased lumbar flattening to $20.3 \pm 37.6\%$ of natural standing lordosis, while the medium and large pads increased lumbar lordosis to $39.0 \pm 30.9\%$ and $57.0 \pm 34.2\%$ of natural standing lordosis relative to SPAD, respectively (Fig. 5).

3.4.2. Cervical angle

A near-significant effect of backrest condition was observed for the cervical angle only when normalized to erect sitting [$F_{4,104} = 3.2$, $p = 0.02$]. The mean cervical angle was measured to be $37.7 \pm 3.2^\circ$ in erect sitting and $34.8 \pm 3.4^\circ$ in natural standing.

3.5. Subjective measures

3.5.1. Body part discomfort questionnaire (BPDQ)

Overall, discomfort values were low, ranging from a mean of 0.2–1.0 for each condition on the 0–5 scale.

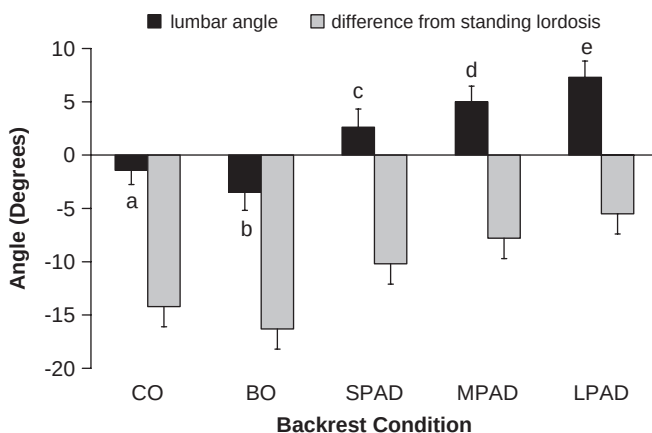


Fig. 5. Lumbar angle as a function of backrest condition and as the difference from natural standing lordosis ($12.8 \pm 1.9^\circ$) (with standard error, gender pooled). Bars with different letters represent significant difference between lumbar angles at $p < 0.02$.

A mixed ANOVA performed on each body part revealed significant effects of backrest condition on discomfort for the middle lower back [$F_{4,112} = 3.4$, $p = 0.01$] and the right upper back [$F_{4,112} = 3.5$, $p = 0.01$], while the left upper back approached significance [$F_{4,112} = 2.8$, $p = 0.03$]. Discomfort in the middle lower back, right and left upper back was increased by the large lumbar pad (LPAD) (Table 3—note only 9 of 13 areas are presented as the popliteal and thigh regions were always rated “0” by each participant).

3.5.2. Backrest condition rating (BCR)

A significant main effect for backrest configuration was found for the backrest rating [$F_{4,104} = 8.4$, $p = 0.00001$] ($N = 28$, two participants did not complete the questionnaire). SPAD was rated significantly higher than all other conditions ($p < 0.02$) except for BO (Fig. 4(d)). Multiple regression analyses revealed no relationships between spine posture, pressure, anthropometrics, and the backrest rating.

3.5.3. Rank-order questionnaire

A significant backrest condition main effect was found for the rank-order questionnaire [$F_{4,112} = 14.8$, $p = 0.000001$]. The SPAD condition was ranked best, followed by BO, MPAD, CO, and LPAD. SPAD was ranked significantly better than all conditions ($p < 0.02$) except for BO, while BO was ranked significantly better than CO and LPAD ($p < 0.02$) (Fig. 4(e)).

4. Discussion

This is one of the first studies to examine the effects of backrest configuration on biomechanical measures and comfort during a typical office word-processing task. The addition of a supplementary backrest to a chair alone decreased both average backrest pressure and MCA by 20% (Figs. 4(a) and (c)), and mean peak backrest pressure by 35% (Fig. 4(b)). Both average and mean peak backrest pressure increased with lumbar pad thickness. In terms of user preference, the backrest alone or with the small lumbar pad were most highly rated (Figs. 4(d) and (e)). These conditions had the lowest MPP and smallest contact area on the backrest as well as minimal (small pad) to no lumbar lordosis (BO) (Fig. 5). Backrest preference was found to be independent of individual size or stature.

We found that backrest configuration had a significant influence on backrest AP, MPP and MCA during word processing (Fig. 3(a)). Although our task differed, a previous study of assembly-line sitting, reading and writing found no difference in pressure measures between a flat and 7 cm thick backrest (Bendix et al., 1996). In the current study, the two most preferred conditions, BO and small (3 cm) pad had significantly lower mean peak backrest pressure than the 6 and 9 cm lumbar pads. An ‘even’ pressure distribution, one without pressure peaks, is commonly suggested as a comfort/discomfort criterion, and is essential to avoid discomfort in specific areas, such

Table 3

Mean body-part discomfort as a function of backrest condition (values in parentheses indicate standard deviation)

Body-part	Backrest condition				
	CO	BO	SPAD	MPAD	LPAD
L. neck	0.35 ^a (0.47)	0.28 ^a (0.54)	0.41 ^a (0.67)	0.41 ^a (0.76)	0.39 ^a (0.78)
R. neck	0.41 ^a (0.61)	0.44 ^a (0.66)	0.44 ^a (0.68)	0.56 ^a (0.89)	0.41 ^a (0.76)
L. upper back	0.54 ^{ab} (0.88)	0.51 ^a (0.89)	0.70 ^{ab} (0.90)	0.62 ^{ab} (0.97)	0.88 ^b (1.22)
R. upper back	0.55 ^a (1.01)	0.60 ^a (0.94)	0.82 ^{ab} (0.98)	0.75 ^{ab} (1.14)	1.0 ^b (1.36)
L. lower back	0.46 ^a (0.76)	0.40 ^a (0.65)	0.49 ^a (0.70)	0.48 ^a (0.72)	0.67 ^a (0.83)
R. Lower Back	0.49 ^a (0.79)	0.51 ^a (0.67)	0.60 ^a (0.69)	0.66 ^a (0.92)	0.82 ^a (0.97)
M. Lower Back	0.67 ^a (0.81)	0.67 ^a (0.82)	0.60 ^a (0.70)	0.76 ^a (0.93)	1.09 ^b (1.01)
L. Buttock	0.44 ^a (0.70)	0.60 ^a (1.04)	0.71 ^a (0.92)	0.66 ^a (1.11)	0.83 ^a (1.29)
R. Buttock	0.46 ^a (0.72)	0.58 ^a (1.03)	0.64 ^a (0.83)	0.74 ^a (1.20)	0.86 ^a (1.31)

Means with different letters (in a row) represent significant difference ($p < 0.02$) between backrest conditions.

as the back and thighs, as concentrated pressure can lead to tissue ischemia (Houle, 1969; Sprigle et al., 1990). That being said, regression analyses indicated that pressure was not a good predictor of preference or discomfort, when controlling for other variables, supporting the car seat research which also has not found a clear relationship (Gyi and Porter, 1999; Porter et al., 2003). Combining lumbar angle and pressure, we found all three lumbar pads promoted lordosis by supporting the lumbar region, corroborating previous research that found that support in the lumbar area lessened discomfort while greater lumbar discomfort was associated with isolated thoracic support or no contact with the backrest (Vergara and Page, 2000a).

The association of lumbar support thickness to discomfort and preference suggests an optimal size of lumbar support between no additional support (BO) and 3 cm (small pad). When lumbar support thickness was increased beyond 3 cm, backrest rating and ranking became progressively worse (Figs. 4(d) and (e)) while middle lumbar discomfort increased. Although the above two conditions were preferred, they did not preserve lumbar lordosis well (only slight lordosis was maintained with the small pad). The large pad was preferred least, and while it best maintained lumbar lordosis, a number of individuals complained that it pushed them forward (resulting in the most anterior CP, Table 2). In terms of comfort, Bishu et al. (1991) reported that an intermediate lumbar angle was preferred, while the current study found no relationship to lumbar posture per se. (Also in contrast to Bishu et al., we found no association between cervical angle and backrest preference or discomfort.) The current findings are interesting given that maintaining lumbar lordosis during sitting has been said to decrease intervertebral disc pressure and ligament tension, both of which have been associated with low back pain (Adams et al., 1980; Adams and Hutton, 1985; Andersson et al., 1975; Hedman and Fernie, 1997). However, the preferred seating conditions (with straighter lumbar spines) may, in some way, reflect improved transport of disc metabolites, associated with a flexed spine, especially if alternated with lordotic postures

(Adams and Hutton, 1985). The preferred amount of lordosis may be related to the pain state of the individual as reduction of back and leg pain was found in low back pain patients using a much larger (13 cm) lumbar roll than the current study (Williams et al., 1991).

While our data may indicate that contact area and peak pressure may be more useful measures in backrest evaluation than AP, it may be that a combination of pressure and posture variables may be even more discriminative. For example, comparing the two most preferred conditions, the backrest by itself had the lowest average and MPP (Fig. 4) along with a straight lumbar spine (Fig. 5) while the small pad had the lowest MCA, low peak pressure (Fig. 4) but maintained some lordosis (Fig. 5).

It may also be that *changes* in posture or position rather than posture itself, is important. Vergara and Page (2002) suggested that large changes in posture are indicative of discomfort while small “micro” movements are necessary to alleviate pain caused by static lumbar and pelvic postures. Consequently, tracking CP motion may prove to be an effective predictor of discomfort and backrest preference (the former suggested by Fenety et al., 2000). In the current study, the addition of the backrest itself allowed the largest variability (although similar to the chair alone) while the addition of the lumbar pads acted to constrain movement of the CP. These data may indicate that seating conditions which allowed (or promoted) more movement were deemed more comfortable.

Personal anthropometrics had no influence on backrest preference, although a previous study found chair and participant anthropometric interactions, leading to a conclusion that anthropometrics were important in chair selection (Bishu et al., 1991). We did find a gender difference, as males had higher average seatpan pressure and greater backrest contact area than females. In the current study males had larger mass, BMI, and WHR than females (Table 1). We did not find a gender effect on posture as found by Dunk and Callaghan (2005), likely due to the use of a backrest in the current study. In spite of incorporating many additional measures in our analyses,

we were unable to improve on the work of Coleman et al. (1998) who found no relationships between anthropometrics (height, gender and BMI) and preferred lumbar support depth.

There are several limitations to this study. First, while our simple and inexpensive lumbar and cervical posture device was deemed effective and did not interfere with pressure measurements or behaviour, the measured lumbar angles were smaller than expected. This was likely due to the length of the sensors (11.4 cm) resulting in less bending than anticipated. It was thought that normalizing the angles to standing posture would assist in interpretation; however, the standing postures were also quite variable. In addition, in some individuals, the CO and BO conditions resulted in “flexed” lumbar angles which must be viewed with caution as the sensors were unidirectional and mounted to measure lordotic angles. Note that it was possible that the sensor did not follow the spine but may have been influenced by the pad or backrest. It must also be noted that, with the addition of each successively larger lumbar pad, the backrest vertical CP was altered due to the placement of the pad under the mat, and is marginally lower than reported in Table 2 for each lumbar pad condition. However, the relationship is complicated by compression of the foam pad. To allow for the thickness of the backrest and ensure that the seatpan depth was not a factor, the seatpan was moved forward 5 cm to allow for the thickness of the supplementary backrest, thus effectively maintaining the same seatpan depth for all conditions (as shown in the CP values). Finally, while it could be argued that EMG may have been beneficial, pilot work indicated that the electrodes interfered with pressure measurement, and the activity levels were too low and variable to delineate changes with backrest configuration, similar to previous studies (Andersson and Ortengren, 1974; Andersson et al., 1974, 1975).

The current study used a 15-min protocol, which was a compromise between very short biomechanically-based studies (less than 1 min) and subjective studies which indicate trials of at least 1 h may be required to distinguish changes in comfort (Bishu et al., 1991; Shackel et al., 1969). The 15-min protocol was shown to produce significant differences in preference and comfort, while at the same time minimizing data collection duration. It is possible that our relatively short collection duration may have missed adaptation to the lumbar pad, and subsequent changes in pressure distribution or posture, that may have occurred over a longer period. It has been suggested that individuals tend to adopt “passive” or slumped postures when seated for a prolonged duration, which may lead to low back pain due to deactivation of the erector spinae muscles thus relying on spinal ligament support, which may lead to injury over time (Adams et al., 1980; Adams and Hutton, 1985; Andersson et al., 1975; Hedman and Fernie, 1997). However, pilot data indicated no change in pressure or EMG over a 30 min trial, thus justifying the 15 min trial time used.

5. Conclusions

This study has provided new insights into the effects of lumbar support on biomechanical variables and comfort in seated office work. The addition of a supplementary backrest significantly reduced average backrest pressure, mean peak backrest pressure, and mean contact area on the backrest. In addition, a 3 cm lumbar pad tested provided lumbar support and prevented flattening of the lumbar spine. Participants preferred the small lumbar pad (3 cm thick) or supplementary backrest only, regardless of anthropometrics, suggesting an optimal amount of lumbar support exists. These conditions also had lower peak backrest pressure, greater seatpan contact area but smaller backrest contact area. Future studies are needed to understand these relationships under longer durations and different seating conditions.

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